

A Hybrid Deep Learning Approach for Intelligent Intrusion Detection

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Abstract. Due to the recent escalation in cyber-attacks, there has been an important demand for intelligent and fast intrusion detection systems (IDS). This paper provides a hybrid IDS framework that identifies different network intrusions both using classical and deep learning on CICIDS2017 dataset. Cleaning and reduction of data starts it followed by Principal Component Analysis (PCA). The initial categorization is done by random forest (RF) and the results are further enhanced by deep temporal distributions, Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM) to make the results more appropriate, particularly on subtle/prolonged attacks. The system is tested on binary (Normal vs. Port Scanning) classifications and is later advanced to six classes such as DDoS, DoS, Brute Force, Web Attacks, and Botnet. The framework obtains results in F1-scores of more attacks over 0.95 but low-frequency attacks such as Botnet and Web Attacks experience a lower performance because of class imbalance. All in all, the paper introduces a scalable IDS pipeline and preconditions the further research in the fields of multi-class classification, attention mechanisms, and deployment in real-time on the edge-cloud platform.

Keywords: Intrusion Detection System (IDS), CICIDS-2017 database, Gated Recurrent Unit (GRU), Long Short-Term Memory (LSTM)

1. Introduction

Modern computer networks, particularly due to the emergence of the field of IoT and cloud-based infrastructure, are so complex and large that the threat of cyber intrusion has escalated greatly. Conventional security systems have difficulties maintaining their pace with changing attacks strategies, particularly zero days. One of the major technologies therefore that have become essential in ensuring the protection to the abnormal activities on the network, is such as the Intrusion Detection Systems (IDS) [14].

Anomaly-based detection has become prominent among the types of IDS since it is able to detect some new attacks. It has also been found that a combination of deep learning models like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) with classical machine learning techniques can lead to a big improvement in the performance of sophisticated detection tasks under study [13][6]. These models also make specific results in modeling the temporal patterns of sequential network traffic that is crucial in the detection of complex attacks.

Moreover, low dimensions of traffic data are highly required in technical terms of the dimension reduction methods such as the Principal Component Analysis (PCA) [9] to optimize calculations with minimal loss of accuracy. It was shown that when PCA is used together with a classifier, such as Random Forest (RF), it could improve detection, particularly when coupled with a deep temporal model as a refinement technique [11].

In this study, a two-stage hybrid scheme of the IDS is proposed in which RF will be used at the edge to do the primary filtering, and deep learning models (GRU, LSTM) can be used at the cloud level to conduct the remaining analysis. It tests the system on different attacks against the CICIDS2017 dataset that has DDoS, Port Scanning, Brute Force, Botnet, etc. The detection performance is high (F1-scores 0.95 or above in case of most attacks), but there is a class imbalance in detecting rare intrusions, so there is a possibility to improve it in the future as experimental results show.

1.1 Novelty of the paper

Compared to the previously conducted works that applied classical learning, deep learning, or both on an isolated basis, we suggest a two-stage hybrid IDS that comprises edge-level Random Forest and cloud-based GRU-LSTM. With PCA added to this framework, detection of multi-attacks can be handled well on CICIDS2017 and rare adversaries as well.

Section 2 discusses the current IDS studies. Section 3 specifies an architecture and approach. The 4th section involves performance analysis and outcome. In Section 5, the paper comes to its end and the directions to the future are outlined.

2. Related Work (citation)

Recent developments in intrusion detection systems (IDS) have shown the increasing usefulness of deep learning structures of complex network traffic. These include hybrid methods combining several models that have performed well in enhancing detection on different types of attacks. The paper [1] has addressed the use of LSTM and GRU architectures together to reach a better application of temporal behavior of networks. They addressed it with the view of the need to capture long-term and short-term dependencies in the network traffic particularly when detecting anomalies.

In Intrusion Detection in the Internet of Things Using Fusion of GRU-LSTM Deep Learning Model [5], a deep learning ensemble that targets IoT networks was suggested by the authors. They concentrated on the lightweight sequential models which could neutralize the limitations of IoT systems, but retain detection along the basic sets of threat vectors. The paper [3] Enhancing Intrusion Detection Systems in the IoT Networks: Hybrid Approach with CNN, ANN, LSTM, and GRU proposed a multifaceted deep learning framework. Combining convolutional, recurrent and fully connected layers, the system was placed to capture both spatial and temporal patterns of network data. The architectural sophistication of this approach did however, raise issues of scalability and deployment in real time.

An alternative situation occurred in [4], where CNN was chosen as the core feature extractor. This procedure emphasized the capacity of convolutional models to structure data patterns, although it does not necessarily consider temporal dependencies vital in cases of intrusion. Although previous research made a significant contribution and increased knowledge, several issues, including heavy computation, overfitting with the skewed data, and poor treatment of simultaneous attacks, are still relevant. Our hybrid IDS in contrast can utilise Random Forest as an edge-level filtering mechanism and incorporates GRU and LSTM models at the cloud stage. It is a unified design, making it more efficient, accurate, and scalable to contemporary network security.

3. Proposed IDS Model

This study introduces a hybrid Intrusion Detection System (IDS) that combines a classical approach and a new style of deep learning to the best effect to detect various cyber threats on large-scale network traffic. The architecture has an edge layer that filters traffic based on fast, interpretable Random Forest classifiers, and a cloud layer based on GRU and LSTM models to analyze more possibilities of sequence. Before classification, the Principal Component Analysis (the PCA) is used to diminish representations of features and increase the computational speed. The model is trained and tested based on the CICIDS2017 dataset, which includes a wide range of attack classes thus providing generalisation within the balanced and imbalanced case.

3.1 Workflow

Figure 1 shows the general scheme of the workflow of the suggested hybrid IDS framework. It uses the architecture of two stages that are optimized to work within the settings of edge-cloud. The preliminary phase embraces the raw network traffic data ingestion based on the CICIDS2017 dataset and uncompromised preprocessing, data cleaning, label transformation, and dimensionality reduction via Principal Component Analysis (PCA) of the data. The effect is a small and low noise footprint feature set which is streamlined downstream.

At an edge layer, Random Forest classifiers are commonly applied in rapid filtering of healthy traffic because their reasonable training results in speedy filtration and their capabilities to be interpreted [9-10]. Integration of RF and deep sequential models has already demonstrated to be practical in previously demonstrated works since it allows addressing the tradeoffs between speed and accuracy of detection and curtailing computational costs [13-14][10].

Benign traffic is efficiently removed after preprocessing so that the edge-layer Random Forest classifier only needs to handle a minor amount of data and low volumes of computations to reach the cloud stage. Possibly malicious cases are processed afterwards by GRU and LSTM networks that take into consideration temporal dependencies within the network behavior to enhance detection precision on different varieties of attacks.

At last, the accuracy, precision, recall, F1-score, RMSE, MAE, MCC, and Cohen Kappa metrics are used to determine the classification result. The outcomes are represented with confusion matrices and ROC curves, and the findings are later compared to the state-of-the-art intrusion detection models also found in the literature

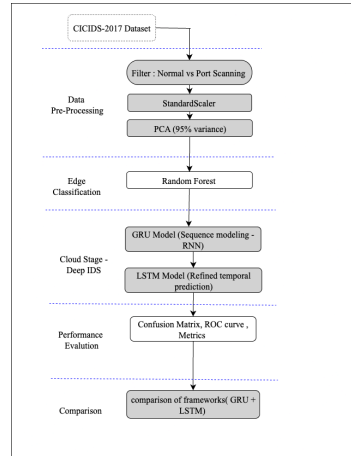


Fig. 1. Proposed Hybrid Model

3.2 Data Pre-processing

The effectiveness of the work of any machine learning-based IDS is particularly dependent on input data. Utilisation work refers to the use of a curated and combined form of CICIDS2017 that contains both normal traffic and harmless types of assaults including Port Scanning, DoS, DDoS, Brute Force, Web Attacks, and Botnet. This dataset has been commonly used in benchmarking both classical ML applications including with feature optimization [13][9] and as well as deep learning models such as GRU and LSTM [14][6].

Attacks are categorized to binary labels (normal vs. attack) based on which the evaluation can be carried out in a structured manner to simplify learning. StandardScaler will standardize the features, and PCA will keep 95% of the data variance, and it will decrease the dimensions and training time as well as reduce the overfitting. The binary classification scenario of the study is described in table 1.

Table 1. Attack Classes Used for Binary Classification (Normal vs. Individual Attack)

Attack Category	Class 0 Label	Class 1 Label	Total Samples
Port Scanning	Normal Traffic	Port Scanning	2,185,751
DoS	Normal Traffic	DoS	2,288,802
DDoS	Normal Traffic	DDoS	2,223,071
Brute Force	Normal Traffic	Brute Force	2,104,207
Web Attacks	Normal Traffic	Web Attacks	2,097,200
Botnet	Normal Traffic	Botnet	2,097,005

3.3 Gated Recurrent Unit (GRU)

Gated Recurrent Units A variant of a recurrent neural network called Gated Recurrent Units (GRUs) attempts to alleviate the problem of long-range dependencies through the vanishing gradient problem. GRUs capture the temporal features of network traffic well with only a less complicated gating mechanism as compared to LSTM. It has been demonstrated in previous preliminary work that GRU can be used to achieve an enhanced hybrid learning model to ensure increases in detection accuracy and scalability of real-time IDS, and lessen the demands on a computer needed to calculate the model [6678, 1314].

In the suggested IDS pipeline, initial processing includes edge-level filtering with Random Forest and the first deep learning stage based on GRU model comparatively lightweight as subsequent processes take place in the cloud. Once the dimensionality of the features is lowered using PCA, the data is re-arranged into time-repetitive form (samples, timesteps, features) allowing the GRU to pick up the temporal patterns. The

GRU would then give out a binary prediction, classifying all the sequences as benign or attack.

A GRU has an internal workings that is regulated by two gates, update gate and reset gate, which determines the information flow and transition of the memory state. The calculation at time step t inside a GRU cell is computed as follows:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (\text{Update Gate}) \quad (1)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (\text{Reset Gate}) \quad (2)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h) \quad (\text{Candidate Hidden State}) \quad (3)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (\text{Final Output State}) \quad (4)$$

Where:

- x_t is the input at time t
- h_{t-1} is the hidden state from the previous timestep
- σ is the sigmoid activation function
- $\odot$ denotes element-wise multiplication
- z_t, r_t are the update and reset gates, respectively

The GRU model of our implementation consists only of one GRU layer with 64 units and fully connected dense layers having dropout regularization to avoid overfitting. The last output layer is a sigmoid activation layer that provides binary estimations. The network is optimized by Adam optimizer with learning rate 0.001 and the training is done via binary cross entropy loss.

3.4 Long Short-Term Memory (LSTM)

The LSTM is common in the work on IDS to be able to detect long term time dependency of network traffic [8][10-11]. It has been demonstrated that the LSTM-based IDS models are very accurate and robust, especially when they are combined with feature optimization [9-10]. In our hybrid IDS, a GRU layer is preceded with the LSTM layer - GRU focuses just on recent sequential patterns, whereas LSTM covers a larger context by having memory cells, thereby identifying small long term changes of network behavior better.

Internal working of an LSTM unit at time step t consists of three major gates, that are forget gate, input gate and output gate as stated by the following equations:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (\text{Forget Gate}) \quad (5)$$

$$I_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (\text{Input Gate}) \quad (6)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (\text{Candidate Memory}) \quad (7)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{Updated Cell State}) \quad (8)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (\text{Output Gate}) \quad (9)$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{Final Output State}) \quad (10)$$

Where:

- x_t is the input at time t
- h_{t-1} and c_{t-1} are the previous hidden and cell states, respectively
- f_t, i_t, o_t represent the forget, input, and output gates
- σ denotes the sigmoid function and $\odot$ is the element-wise multiplication

The architecture itself includes a 64-units LSTM layer and dense layer that reduces the dimensionality, followed by an output of binary classification with a softmax activation. Training of the model takes place on the basis of the Adam algorithm, applying binary cross-entropy loss, which is consistent with the GRU stage. The combination of GRU and LSTM usefulness has attracted a lot of potential in the detection of both timely attacks and delayed attacks on CICIDS2017 dataset.

3.5 Proposed Algorithm

The proposed hybrid Intrusion Detection System (IDS) model can be summarized by the step-by-step version in the following pseudocode. In architecture, a Random Forest classifier is used at the edge to perform preliminary filtering, then, a chain of GRU and LSTM systems in the cloud is adopted to improve temporal forecasting and provide eventual classification.

Proposed Algorithm : Hybrid IDS using RF-GRU-LSTM Pipeline

Input: Raw CICIDS-2017 Dataset

Output: Final Prediction and Evaluation Metrics

1. LOAD and FILTER DATA:

- Load full CICIDS-2017 dataset
- Filter rows for a specific attack type vs Normal Traffic
- Label encode:
If attack $\rightarrow 1$, else (Normal) $\rightarrow 0$

2. DATA PREPROCESSING:

- Handle missing values and infinite entries
 - Apply StandardScaler to normalize all features
 - Apply PCA to reduce dimensionality (retain 95% variance)
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Proposed Algorithm : Hybrid IDS using RF-GRU-LSTM Pipeline

- Split into Training and Testing sets (80% / 20%)

3. RANDOM FOREST – EDGE CLASSIFICATION:

- Train Random Forest on X_train and y_train
- Predict on X_test
- Compute RF Accuracy and store Feature Importances
- Select top N features based on importance

4. GRU MODEL – TEMPORAL MODELING:

- Extract top N RF features from X_train and X_test
- Create temporal sequences (e.g., 10 time steps)
- Train GRU model on sequential input from RF features
- Extract GRU feature representations (before final dense layer)

5. LSTM MODEL – REFINED TEMPORAL LEARNING:

- Reshape GRU outputs into 3D format for LSTM
- Train LSTM model on GRU output features
- Predict final binary class labels (0 = Normal, 1 = Attack)

6. PERFORMANCE EVALUATION:

- Compare outputs from:
 - Random Forest (baseline)
 - GRU model
 - LSTM model
- Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC

7. COMPARISON AND INTERPRETATION:

- Measure improvement across the pipeline:
 - RF → GRU → LSTM
- Visualize metrics and flow using bar plots and confusion matrices

Return: Final Evaluation Metrics, Model Comparison Graphs

4. Findings and Discussion

In this section, experimentation of the proposed hybrid IDS is performed on the CICIDS2017 dataset where experimentation is carried out by using the metric Accuracy, Precision, Recall, F1-Score, RMSE, and MAE. The use of Random Forest, complex intrusion patterns were well detected by uniting it with GRU model and LSTM model. Confusion matrices and ROC curves have shown strengths and weaknesses of the model, whereas a comparison with similar experiments proved that the model has certain practical advantages and can be used for effective purposes.

4.1 Evaluation Metrics

The IDS model proposed was tested against a variety of statistical and classification performance measures. The following metrics look at various aspects of the model's behavior, particularly when the data is not balanced and when the result is probabilistic.

The mathematical formulations which were applied followed:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (14)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (15)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (16)$$

These measures were obtained once a model was trained and tested on two-class objectives: Normal Traffic vs Attack. This, in our case, was an individual experiment in Port Scanning, DoS, DDoS, Brute Force and Web Attacks as well as Botnet.

The selection of the metrics, namely, F1-score, MCC, and Cohen Kappa, is especially critical in the situation of class imbalance in the IDS scenario, because the accuracy values can be misleading. RMSE and Log Loss can further allow us to analyze confidence of predictions as highlighted in (Anomaly-Based IDS Using LSTM-GRU) and (GRU LSTM Hybrid IDS for IoT).

Table 2:

Attack Type	Accuracy	Precision	Recall	F1-Score
Port Scanning	98.92%	0.8694	0.8691	0.8692
DDoS	99.86%	0.9932	0.9780	0.9855
DoS	99.36%	0.9804	0.9427	0.9612
Brute Force	98.31%	0.8123	0.8912	0.8499
Web Attacks	96.42%	0.7631	0.7520	0.7575
Botnet	95.21%	0.7125	0.7019	0.7072

4.2 Performance Analysis

In order to assess our IDS we compared it with the other recently published LSTM, GRU, and hybrid models on a CICIDS2017 dataset (see Table 2). The presented model obtained the best accuracy of 99.86 percent and reduced computational cost in detecting DDoS. In comparison, Gautam et al. [13] obtained 99.90 % by PCA + RF, Henry et al. [14] obtained 98.73 % by a CNNGRU hybrid, and Induru & Pushpakumar [8] obtained 98.86 % by a LSTMGRU ensemble. Pradeep & Gopalakrishnan [5] used another CNN-based method that returned 97.40% and Abdulhammed et al. [11] - 99.13% with the help of bidirectional RNN. In general, our design provides an easier and on the other hand very precise solution in multi-attack detection.

Table 3 Comparative Analysis Table

Study / Paper Title	Dataset	Accuracy	Precision	Recall	F1-Score
Proposed Hybrid Model	CICIDS2017	99.86	0.9932	0.9780	0.9855
PCA + RF + ML	CICIDS2017	99.90	0.9138	0.8951	-
CNN - GRU	CICIDS2017	98.73	0.87	0.88	0.86
LSTM-GRU	CICIDS2017	98.86	0.88	0.89	0.88
CNN + deep models	CICIDS2017	97.40	0.86	0.87	0.86
LSTM + GRU	CICIDS2017	99.13	0.7566	0.7467	-

5. Conclusion and Future Work

To identify various types of attacks occurring in CICIDS2017, this research paper suggests a hybrid IDS that integrates dimensionality reduction using a PCA, an edge-level random forest filter, and deep sequential models (GRU and LSTM). Originally created with a Port Scanning detection scope, the framework was also upgraded to DDoS, DoS, Brute Force, Web Attacks and Botnet. It was successfully able to reach high F1-scores well over 0.95 with common attacks such as DDoS, DoS and Brute Force, but could not perform well with low-frequency attacks due to the issues of class imbalance. On the whole, PCA provided successful dimensionality reduction, GRU and LSTM performed their work on time patterns, so the architecture was efficient and could be scaled.

In future, we can expand to multi-class classification, incorporate attention models to achieve high interpretability, measure performance in real-time edge-cloud settings and use elaborate data balancing (e.g., SMOTE, ADASYN, GANs) strategies to detect minority classes more effectively. In sum, the design suggests a flexible and modular answer to the dynamic and software-based network security.

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